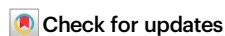


AI allows pre-screening of *FGFR3* mutational status using routine histology slides of muscle-invasive bladder cancer

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Pathogenic activating mutations in the fibroblast growth factor receptor 3 (*FGFR3*) drive disease maintenance and progression in urothelial cancer. 10–15% of muscle-invasive and metastatic urothelial cancer (MIBC/mUC) are *FGFR3*-mutant. Selective targeting of *FGFR3* hotspot mutations with tyrosine kinase inhibitors (e.g., erdafitinib) is approved for mUC and requires *FGFR3* mutational testing. However, current testing assays (polymerase chain reaction or next-generation sequencing) necessitate high tissue quality, have long turnover time, and are expensive. To overcome these limitations, we develop a deep-learning model that detects *FGFR3* mutations using routine hematoxylin-eosin slides. Encompassing 1222 cases, our study is a large-scale validation of a model prescreening *FGFR3* mutations for MIBC and mUC patients. In this work, we demonstrate that our model achieves high sensitivity (>93%) on advanced and metastatic cases while reducing molecular testing by 40% on average, thereby offering a cost-effective and rapid pre-screening tool for identifying patients eligible for *FGFR3* targeted therapies.

Urothelial carcinoma (UC), mainly localized in the bladder, is a common malignancy in women and is the fourth most common malignancy in men¹. At diagnosis, over 75% of patients show a non-muscle-invasive bladder cancer (NMIBC) that can be successfully managed with local bladder preserving surgical treatment, instillation therapy in selected risk groups, and surveillance. The remaining 25% of patients exhibit a muscle-invasive disease (MIBC), which usually requires aggressive treatment (cystectomy, radiotherapy, or palliative treatment). Despite

progress in understanding the disease and therapeutic strategies, survival rates of locally advanced and metastatic bladder cancer are still poor². However, very recent major therapeutic improvements such as enfortumab vedotin plus pembrolizumab for the treatment of metastatic urothelial carcinoma raise the hope that the outcome rates of urothelial carcinoma patients can be significantly improved in the near future³.

One of the central biological pathways driving NMIBC involves the fibroblast growth factor receptor 3 (*FGFR3*). *FGFR3* encodes a cell

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surface receptor that plays a pivotal role in cell proliferation, differentiation, and survival. Pathogenic activating mutations, fusions, and amplifications of the *FGFR3* gene (fusions are rare) have been extensively linked to the development and maintenance of NMIBC, where alteration frequencies range between up to 50% (in stroma-invasive pT1 carcinomas) and up to 80% in non-invasive papillary low-grade urothelial carcinoma, while *FGFR3* alterations are less observed in advanced stage MIBC (non-metastatic and metastatic) or urothelial carcinoma in situ (alteration frequency ranging from 0 to up to 15%; average distribution in nmMIBC and mMIBC around 10 to 15%)^{4–7}. The differences in mutation frequencies can be explained by the fact that bladder cancer is a highly heterogeneous disease on molecular and pathological levels⁸. While papillary non-invasive UC and a relevant subset of stroma-invasive UC mainly arise via a hyperplasia sequence driven by activating *FGFR3* alterations (particularly mutations, but also amplifications and activating gene fusions), the majority of muscle-invasive and relevant subset of aggressive stroma-invasive metastatic UCs arise via the flat carcinoma in situ sequence that is mainly driven by chromosomal aberrations and inactivating *TP53* mutations⁸. This explains the low frequency of *FGFR3* alterations in MIBC and mUC, where *FGFR3* mutant tumors can often be backtracked to long-lasting high-grade non-invasive or stroma-invasive carcinomas progressing to muscle-invasive and subsequently metastatic disease⁸. This illustrates that bladder cancer is a heterogeneous disease, where selective targeting of tumor-driving *FGFR3* alterations—mainly activating hotspot mutations and gene fusions—represents an attractive therapeutic strategy.

Based on these observations, several pharmaceutical companies developed more or less specific *FGFR*-inhibiting small molecules, where erdafitinib is the only FDA and EMA approved for the treatment of *FGFR3* mutant or fused (activating hotspot mutations or/and activating gene fusions) metastatic urothelial cancer after progression on first-line chemotherapy (FDA) or monotherapy with immune checkpoint inhibitors (EMA)⁹. Testing for *FGFR3* mutations and fusions is currently prescribed to identify suitable patients for erdafitinib treatment. It can be performed either with simple hotspot targeting polymerase chain reaction protocols, next-generation sequencing (NGS) on DNA and RNA level, or by specific RT-qPCR assays covering *FGFR2/3* mutations and fusions in a one-shot assay requiring tumor RNA (Therascreen Assay, Qiagen; companion diagnostic assay for erdafitinib)⁹. However, these assays are dependent on high tissue quality (especially those based on RNA), are accompanied by long turnover times (e.g., NGS-based assays), and are relatively expensive if only one candidate gene has to be tested. Thus, the development of pre-screening tools that can be easily and quickly applied in routine diagnostic material, such as hematoxylin and eosin (H&E) glass slides of tumor tissue utilized for routine pathological diagnostics, are of significant interest. This is all the more critical as recently published data from the THOR trials confirmed the clinical efficacy and benefit of erdafitinib in *FGFR3* mutant or fused metastatic urothelial cancer patients¹⁰. In particular, the rapid administration of a further therapy regimen is crucial to increase the likelihood of disease control or therapy response in patient subsets that have already progressed to multiple prior therapy regimens. Thus, pre-screening tools could identify patients with a high probability of having a vulnerable alteration in a fast and cost-effective manner. In contrast, patients with a very low probability could be referred directly to another potentially effective therapy without wasting valuable time on extensive testing.

In this work, we propose a deep-learning model designed to detect *FGFR3*-mutant tumors using routine histology slides (H&E staining) from bladder carcinoma patients. We find that using our pre-screening tool can save on average 40% of molecular tests by ruling out wild-type cases. We address key questions for use in clinical routine by studying our model's confidence in its predictions, inter-block variability, and its performance on histological subtypes.

Results

The model saves on average 40% of molecular tests

The model was trained and validated using a total of 1222 cases across multiple cohorts (Fig. 1a, b). The model's training phase utilized a discovery cohort of 391 cases, achieving a cross-validation area under the receiver operating characteristic (AUC) of 0.82 [0.74–0.90]. External validation was conducted on three independent cohorts of MIBC cases: TCGA MIBC ($n = 307$), MIBC II ($n = 183$) and mUC ($n = 96$). The model was successfully validated with AUCs of 0.82 [0.75–0.88], 0.89 [0.82–0.93], and 0.82 [0.68–0.94] on TCGA MIBC, MIBC II, and mUC, respectively (Fig. 2a–c).

Using an operating threshold calibrated to minimize false negatives, our model demonstrated excellent performance as a pre-screening test across the TCGA, MIBC II, and mUC cohorts. The model achieved near-perfect negative predictive values (NPV) of 0.99 [0.98–1.00] for TCGA, 0.99 [0.98–1.00] for MIBC II and 1.00 [0.97–1.00] for mUC. This high NPV allowed us to reliably identify patients who do not require further molecular testing, potentially saving 144 (47.2%), 72 (37.9%), and 35 (36.5%) molecular analyses in the TCGA, MIBC II, and mUC cohorts, respectively (Table 1). The model's performance across cohorts was as follows (Sensitivity/Specificity/PPV): TCGA = 0.96 [0.92–1.00]/0.47 [0.30–0.72]/0.14 [0.09–0.23], MIBC II = 0.96 [0.92–1.00] / 0.47 [0.36–0.85]/0.21 [0.13–0.45], mUC = 1.00 [0.92–1.00]/0.32 [0.12–0.70]/0.17 [0.10–0.34]. The detailed results can be found in Supplementary Table 2. When evaluated on the identical TCGA cases provided by Loeffler et al.¹¹, our model achieved an AUC of 0.83 [0.76–0.88] in external validation. This represents a substantial improvement of their reported AUC of 0.70 in cross-validation.

Interestingly, we found that using a model trained at a resolution of 1.0 micrometer per pixel (MPP) yielded a performance improvement over models trained at 0.5 MPP (see Supplementary Table 4 for a detailed comparison).

Since our cohorts featured various hotspot mutations (Supplementary Table 7), we investigated whether some hotspots were easier to detect than others. We looked at all the hotspots with more than five occurrences in our dataset and found six: Y373C, S249C, R248C, G380R, G370C, and S371C. S249C was the easiest to detect across the TCGA, mUC, and MIBC II cohorts (Fig. 3a–c).

In our analysis, we evaluated the prevalence of *FGFR3* mutations and the model's prediction accuracy based on sex. No statistically significant differences in *FGFR3* mutation prevalence were observed between males and females across training and validation cohorts (Supplementary Table 15). When stratifying according to sex, the small sample sizes preclude definitive conclusions.

Training on NMIBC cases improves the model performance

NMIBC and MIBC are biologically and pathologically different disease entities with different rates of *FGFR3* alterations. Interestingly, despite known morphological and molecular differences, we observed that a model trained on additional NMIBC cases (MIBC and NMIBC) outperformed a model trained on MIBC cases only (see Supplementary Table 3). The model trained on both MIBC and NMIBC demonstrated superior performance across all external validation cohorts compared to the model trained exclusively on MIBC. Notably, incorporating NMIBC cases in the training set led to significant improvements not only for TCGA NMIBC but also for TCGA MIBC. The model trained jointly on NMIBC and MIBC cases achieved AUCs of 0.87 [0.83–0.91] for TCGA NMIBC and 0.82 [0.75–0.88] for TCGA MIBC, both marking substantial enhancements over the baseline model trained solely on MIBC cases.

Since the TCGA database contained both MIBC ($n = 307$) and NMIBC ($n = 65$) cases, we also validated our model on the pooled cohort MIBC and NMIBC cases ($n = 372$). Our model performed better with an AUC of 0.87 [0.83–0.91]. To further investigate the model's transferability from MIBC to NMIBC, we conducted an external

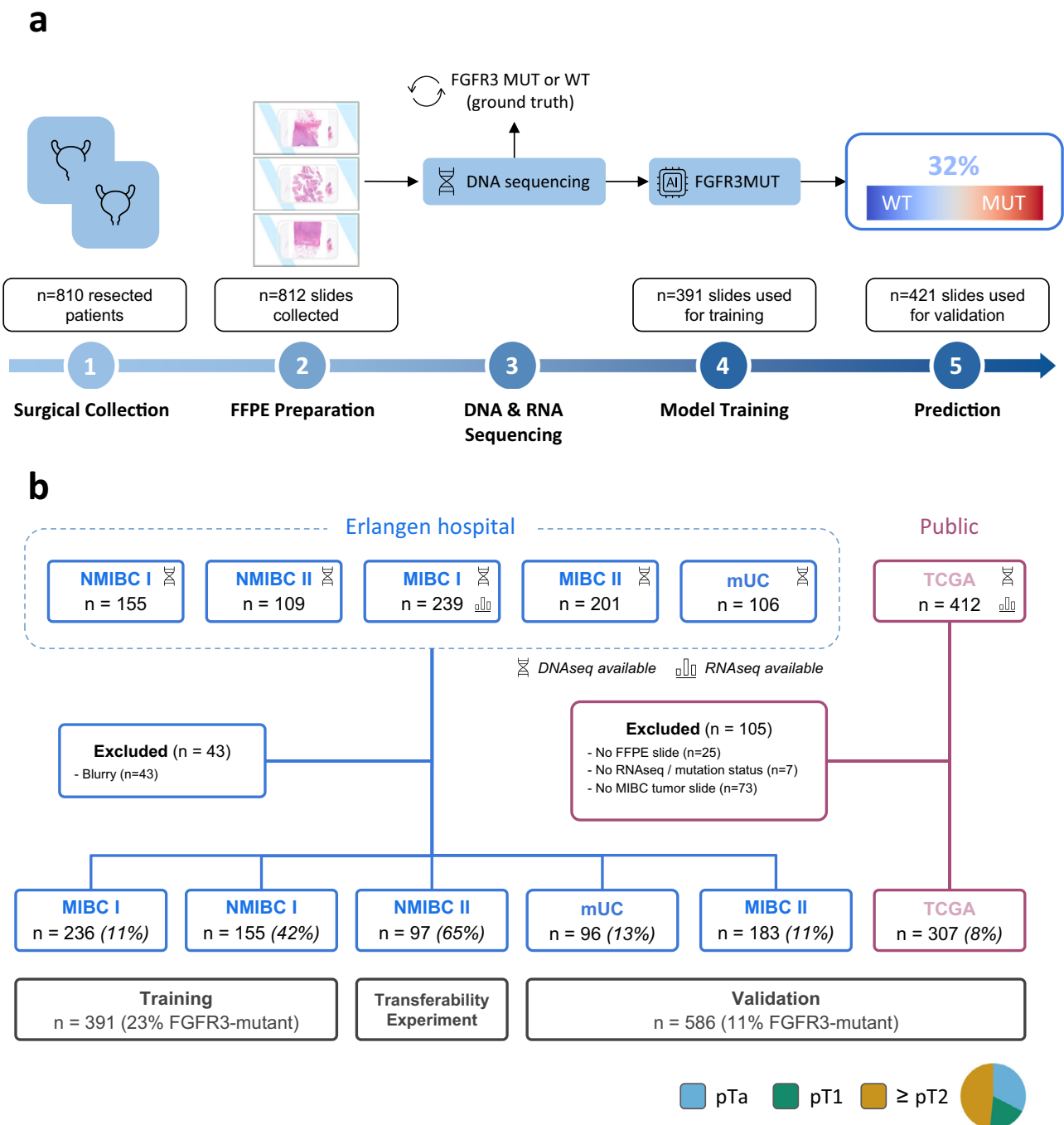


Fig. 1 | Clinical workflow and cohorts presentation. a Multiple cohorts encompassing 810 cases were selected from the Erlangen hospital, where patients had undergone surgical bladder tissue extraction (Step 1). These tissue specimens were then preserved using formalin fixation and paraffin embedding (FFPE) and subsequently stained with hematoxylin and eosin (H&E) for histological examination (Step 2). Following this, consecutive cuts of the tissue were subjected to DNA sequencing utilizing the TSO 500 assay. The DNA sequences were aligned on a reference genome to detect the mutation hotspots across samples. This mapping process enabled the determination of the FGFR3 mutation status among the patients (Step 3). Subsequently, the H&E slides and the FGFR3 mutation status were used as input to train and validate a deep-learning model. **b** FFPE slides from all cohorts were visually inspected to detect artifacts. Slides that did not pass the visual inspection were excluded from the training and validation cohorts. The

model was trained on 391 cases and validated on 586 cases, spanning locally advanced and metastatic muscle-invasive bladder cancer (MIBC) cases. The percentages in *italic* correspond to the final proportion of FGFR3-mutant cases in the cohorts. We made sure that the training and validation cohorts were enriched in MIBC cases (MIBC I for training and MIBC II for validation). A detailed flow chart to select MIBC cases from the TCGA cohort is available in Supplementary Fig. 2. Besides, to explore the model's potential applicability to non-muscle-invasive bladder cancer (NMIBC), we included an additional cohort of 109 NMIBC cases (NMIBC II). While not part of our formal external validation, this cohort served as an exploratory dataset to test the model's transferability. A succinct description of the stages of our cohorts is presented in the bottom right pie chart.

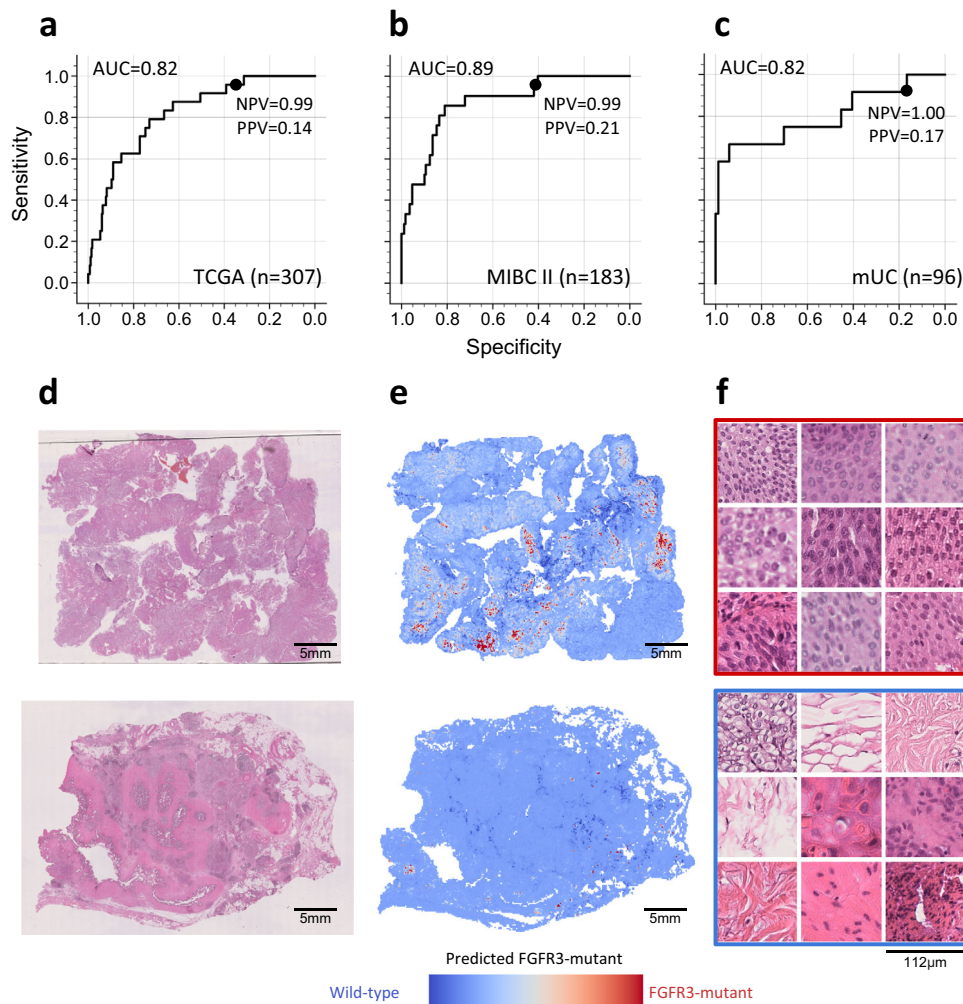


Fig. 2 | FGFR3 mutation prediction. **a–c** Receiver operating curves (ROC) of the FGFR3MUT model on TCGA MIBC (left), MIBC II (middle), and mUC (right). The area under the ROC curve (AUC) is displayed in the upper-left corner. **d** Thumbnail of a FGFR3-mutant case (top) and wild-type case (bottom). **e** The FGFR3 mutation prediction heatmaps showing a probability of each $112 \times 112 \mu\text{m}$ tile harboring FGFR3-mutated features, for a FGFR3-mutant case (top), and wild-type case (bottom). **f** Corresponding most predictive regions of the FGFR3 mutation status (top)

and wild-type status (bottom) across the mUC cohort. We inspected the 100 most predictive tiles of the slide and sampled 9 representative tiles. Regions predictive of FGFR3 mutations revealed monomorphous conventional urothelial tumor morphology and low desmoplastic tumor stroma content. Conversely, regions predictive of the wild-type status contained pleomorphic tumor morphology and desmoplastic tumor content. Source data are provided as a Source Data file.

validation on the cohort NMIBC II (n=97). The model performed worse in this external NMIBC cohort than on external MIBC cohorts, scoring an AUC of 0.70 [0.61–0.80], with sensitivity of 0.93 [0.92–0.97], Specificity of 0.34 [0.13–0.57], positive predictive value (PPV) of 0.73 [0.63–0.82] and NPV of 0.73 [0.50–0.86]. It's important to note that this transferability experiment from MIBC to NMIBC was conducted primarily to explore the model's generalization capabilities. The observed decrease in performance on the NMIBC cohort is not entirely unexpected, as the model was specifically trained and optimized for MIBC cases.

The model performed consistently across histological subtypes

We examined the model's performance across various histological subtypes, categorizing the 3 validation cohorts into conventional urothelial, divergent subtype, and neuroendocrine groups. The distribution of these cases is presented in Supplementary Table 9. Our analysis revealed consistent model performance across these subtypes (Supplementary Table 12). Additionally, we assessed the model's performance in relation to different growth patterns, classifying cases as either solid or papillary (Supplementary Table 10). The results demonstrated stable performance for both patterns

(Supplementary Table 13), with AUC values ranging from 0.84 to 1.00 on MIBC II, 0.81 to 1.00 on mUC, and 0.79 on TCGA MIBC. Lastly, we investigated whether the presence of carcinoma in situ (CIS) affected the model's performance (Supplementary Table 11). This analysis also showed consistent results (Supplementary Table 14), with an AUC of 0.90 [0.79–0.97], 0.89 [0.73–0.99] for cases presenting CIS and an AUC of 0.88 [0.78–0.96], 0.83 [0.61–1.00] with no CIS, respectively on MIBC II and mUC. In all three analyses, the model maintained robust performance across the various categories examined. Besides, we investigated whether *FGFR3*-mutant low-grade or high-grade carcinomas were particularly harder to predict. We found that our model performed better on low-grade cases with an AUC of 0.89 [0.75–1.00] compared to an AUC of 0.82 [0.74–0.88] for high-grade cases on TCGA MIBC.

The model demonstrates high confidence in its predictions

To test the model's robustness, we investigated the variability in prediction scores between slides from the same case (Fig. 4a, b). The intraclass correlation coefficient between the prediction scores from multiple slides on the mUC cohort was 0.66 [0.42–0.90] and 0.89 [0.76–0.96] on the TCGA MIBC cohort. Additionally, given the model's

notably strong performance on the mUC cohort, we sought to explore the differences in *FGFR3* mutation score produced by the model for primary versus metastatic sites. Of the 96 metastatic cases, 4 had at least one tissue sample from the primary tumor site and one from the metastasis. For these 4 cases, we found that none of the slides, whether from the primary or metastatic sites, would lead to a different diagnosis since all scores were significantly higher than the operating decision threshold. Furthermore, to qualitatively assess the difference between primary and metastatic slides, we looked at the top predictive tiles from primary and metastatic sites (Fig. 5b). After careful inspection by a pathologist, no significant differences in histomorphological patterns were found between both sites.

To evaluate the model's confidence in its predictions, we analyzed the variance in predicted class probabilities across the 125 multiple-instance models comprising the final ensemble (see Uncertainty quantification and conformal prediction in the Methods section). We sampled 30 cases each from TCGA (Fig. 4c), MIBC II (Fig. 4d), and mUC (Fig. 4e) cohorts, examining average predictions and their associated variances. Notably, for all observed *FGFR3*-mutant cases across the three external validation cohorts, the prediction variance near the calibrated classification thresholds (Table 1) is low, indicating high agreement among the models in the ensemble in assigning probabilities for these cases. This demonstrates the model's high confidence in identifying *FGFR3*-mutant cases, effectively minimizing false negatives. We further derived conformalized predictions¹² and prediction sets from the model. These 95% confidence prediction sets contain either a single class (wild-type or *FGFR3*-mutant) or both classes when the model lacks confidence. The expected cardinality of prediction sets was 1.42 [1.00–1.71] for TCGA, 1.12 [1.05–1.29] for MIBC II and 1.19 [1.05–1.28] for mUC, showing that for the majority of cases

the model is confident enough to only include one class in its prediction sets.

Histopathology shows patterns linked to *FGFR3* mutations
To ensure our model accurately identified *FGFR3* mutations in pathologically relevant regions, we presented the highest and lowest-scoring predictive tiles to a pathologist for verification (Fig. 2d, e; see Interpretability analysis in the Methods section). The regions most associated with *FGFR3* mutations revealed monomorphous conventional urothelial tumor morphology and low desmoplastic tumor stroma content (Fig. 2f). In contrast, the least associated regions contained pleomorphic tumor morphology and high amounts of desmoplastic tumor stroma (content). These findings align with those reported by Loeffler et al.¹¹, specifically in relation to the predictive accuracy of papillary tumor growth and the presence of homogeneous, regularly shaped nuclei in the most predictive regions of the *FGFR3*-mutation status.

To accurately quantify which histology patterns were linked to *FGFR3* mutations, we selected 400 tiles from TCGA MIBC associated with the *FGFR3*-mutant status ($n=200$) and the wild-type status ($n=200$). A pathologist (M.E.) annotated the regions (tiles), assigning one or multiple histology patterns to every tile (more details can be found in the Supplementary Methods section). We found that most tiles predictive of wild-type and *FGFR3* statuses contained tumor cells, with 99% of mutant tiles and 57% of wild-type tiles containing tumor cells. Tiles predictive of the *FGFR3*-mutant status had a monomorphic appearance, which was observed in 18% of mutant tiles compared to only 4% of wild-type tiles ($p < 0.001$). Tiles predictive of the wild-type status were associated with desmoplastic stroma in the tumor, present in 12% of wild-type tiles but only 1% of mutant tiles ($p < 0.001$). Additionally, inflammation lymphocytes in the microenvironment were observed in 12% of wild-type tiles compared to just 1% of mutant tiles ($p < 0.001$) (Fig. 5a). The detailed findings are presented in Supplementary Table 8.

We also examined the histology patterns associated with the mutation hotspots. We found that the top predictive tiles for *FGFR3* mutations, except for those for G380R, predominantly reflected tumor morphologies with low or absent stroma content and monomorphic and bland-appearing tumor cell nuclei (Supplementary Fig. 1).

Low-expressor *FGFR3* mutants mimic wild-type morphology
Given the prevalent association between *FGFR3* mutations and elevated *FGFR3* expression levels¹³, we carried out a validation excluding atypical *FGFR3*-mutant cases characterized by low *FGFR3* RNAseq expression. We analyzed *FGFR3* RNAseq TPM-normalized expression

Table 1 | Performance of the *FGFR3*MUT prediction model across three external validation cohorts

Cohorts	Predicted <i>FGFR3</i> -mutant patients (%)	Potentially saved molecular tests(%)	Operating threshold (between 0 and 1)
TCGA MIBC	161 (53)	144 (47)	0.059
MIBC II	118 (63)	72 (37)	0.051
mUC	61 (64)	35 (36)	0.063

Predicted *FGFR3*-mutant patients indicate the number of patients the model predicted to have an *FGFR3* mutation. Potentially saved molecular tests represent the number and percentage of patients in each cohort for whom molecular testing could potentially be avoided based on the model's predictions. Operating threshold denotes the classification threshold (between 0 and 1) used for the model in each cohort to achieve these results.

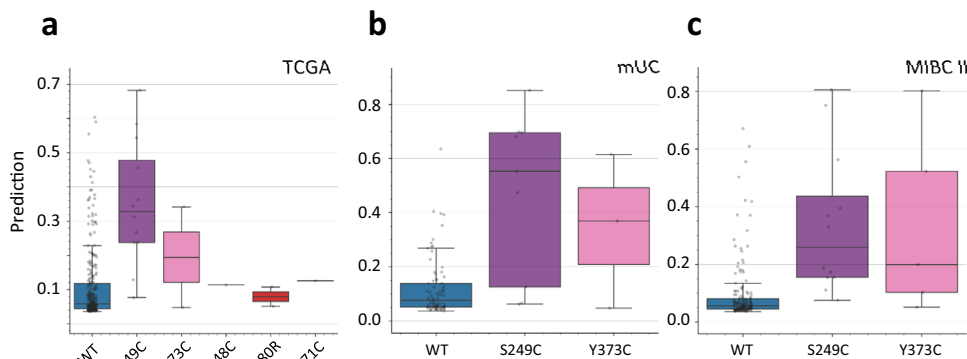


Fig. 3 | Model performance across mutation hotspots. a–c Distribution of model prediction scores for the most prevalent *FGFR3* mutation hotspots in TCGA ($n=300$), mUC ($n=94$) and MIBC II ($n=186$). The wild-type (WT) category is utilized as a reference point for comparison. The exact number of hotspot mutations

in each cohort is given in Supplementary Table 7. Center corresponds to the median, lower and upper hingers indicate the first and third quartiles, and whiskers to the high/lowest value no further than $1.5 \times$ inter-quartile range. Source data are provided as a Source Data file.

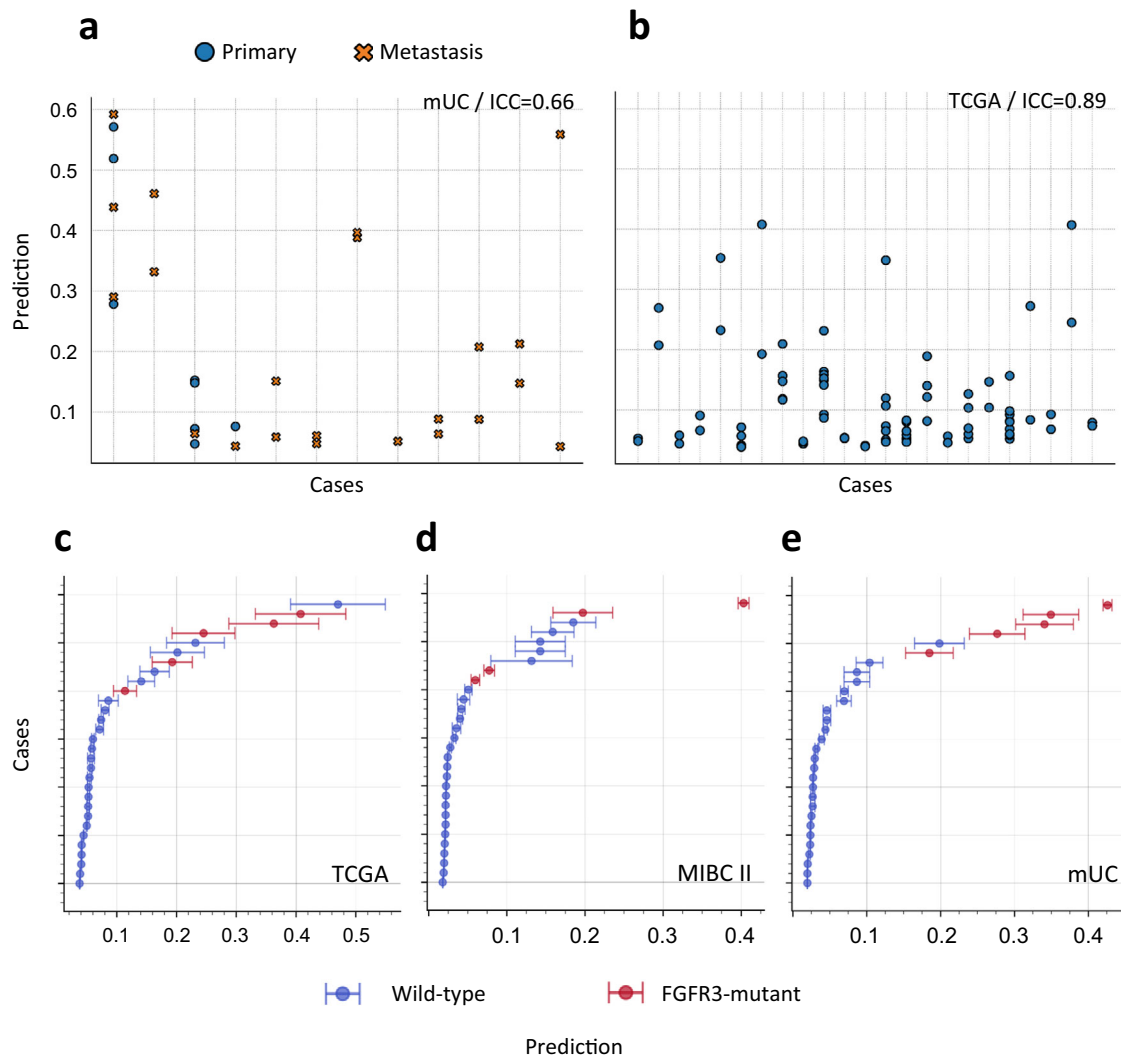


Fig. 4 | Prediction robustness and uncertainty quantification. **a–b** Variability in model predictions across multiple slides from the same patient in the mUC (left) and TCGA (right) cohorts. The intraclass correlation coefficient (ICC) was 0.66 [0.42–0.90] for mUC and 0.89 [0.76–0.96] for TCGA. Metastatic slides are represented by crosses, while primary slides are shown as circles. **c–e** The ensemble model's prediction variability for the FGFR3 mutation status was evaluated by

randomly selecting 30 samples from each validation cohort: TCGA MIBC (c), MIBC II (d), and mUC (e). The data is represented as mean values $\pm$ variance of the predictions, with blue representing true wild-type cases and red indicating true FGFR3-mutant cases. The variances were quantified using deep ensemble uncertainty quantification, and the statistics for each sample were obtained from 150 data points (models) per sample. Source data are provided as a Source Data file.

across the TCGA cohort, identifying the first and third quartiles of expression. Our initial step involved excluding *FGFR3*-mutant cases that were within the lowest quartile of *FGFR3* expression ($n=14$), resulting in an improved AUC of 0.86 [0.79–0.92]. Next, we excluded wild-type cases ($n=82$) that exhibited high *FGFR3* expression, falling into the highest expression quartile, leading to an AUC of 0.85 [0.78–0.91]. By eliminating both atypical groups from our analysis, the model's performance on the TCGA cohort was notably enhanced, achieving an AUC of 0.89 [0.83–0.94] (Supplementary Tables 5 and 6). Our results suggest that mutants with low *FGFR3* (respectively, wild types with high *FGFR3*) expression have a morphology similar to that of wild types (respectively mutants), while *FGFR3* mutant tumors with activating mutations and high *FGFR3* expression are morphologically different. In the TCGA MIBC cohort, we evaluated the mean *FGFR3* log TPM-normalized RNAseq expression between wild-type tumors correctly classified by the model (true negatives; $n=138$) and misclassified wild-type tumors (false positives; $n=144$). The average *FGFR3* expression in true negative tumors was 3.26, compared to 4.28 in false positive tumors, with the difference being statistically significant (Mann–Whitney test, $p=7.9\text{e-}8$).

Discussion

The recent approval of erdafitinib, a tyrosine kinase inhibitor targeting the FGFR signaling pathway, represents the first targeted therapy for metastatic urothelial cancer (mUC) with FGFR alterations¹⁰. Other molecules are currently tested in clinical trials and may represent new alternatives in the coming years in MIBC and NMIBC, with intravesical erdafitinib delivery using the TAR-210 system (NCT05316155). Activating *FGFR3* mutations are recurrent oncogenic mutations in UC. Their frequency is notably high (50–80%) in non-muscle-invasive bladder cancer (NMIBC) but decreases to 10–15% in locally advanced and metastatic UC, with gene fusions being exceedingly rare (~1%)^{4–8}. Both the FDA and EMA approval are bound to proven activating *FGFR3* mutations or gene fusions¹⁰, which can be assessed by the companion diagnostic Therascreen assay or by DNA and RNA-based conventional sequencing or NGS approaches¹⁴. However, all these methods are either expensive or bound to mRNA (Therascreen, RNA-seq), whose quality can be severely altered in FFPE tissues. Given the low frequency in metastatic urothelial cancer, the question arises whether pre-screening tools utilizing existing H&E tumor slides might help identify tumors with a high likelihood of harboring targetable activating *FGFR3*

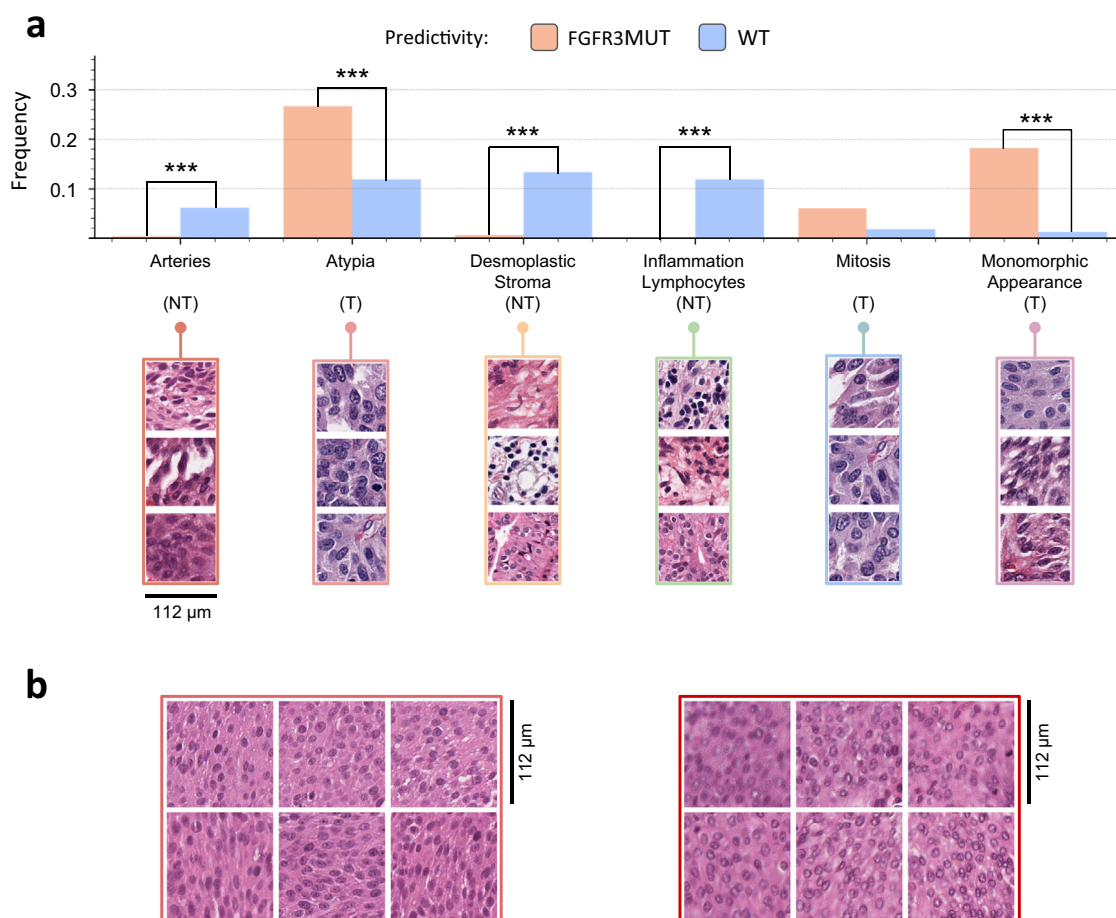


Fig. 5 | Histopathological features associated with FGFR3 mutations.

a Proportion of histology patterns associated with FGFR3 mutation status. A pathologist (M.E.) independently examined 400 representative tiles from TCGA MIBC, equally divided between FGFR3-mutant ($n = 200$) and wild-type ($n = 200$), to identify patterns indicative of each status. Tumor-associated patterns are denoted by (T), whereas patterns identified in non-tumorous regions are represented by (NT). The detailed findings are presented in Supplementary Tables 8 and 9. Key observations include a monomorphic appearance (FGFR3MUT: 18%, WT: 1%, $P = 4.3 \times 10^{-13}$) and atypia (FGFR3MUT: 27%, WT: 12%, $P = 5.0 \times 10^{-3}$) in FGFR3-mutant tumors, while wild-type tumors typically elicit an immune response (FGFR3MUT:

0%, WT: 12%, $P = 2.9 \times 10^{-8}$) and are characterized by the presence of desmoplastic stroma (FGFR3MUT: 1%, WT: 14%, $P = 9.8 \times 10^{-13}$). Statistical significance was assessed using a two-sided z-test for proportions, with Bonferroni adjustment for multiple comparisons. Statistically significant differences were marked with the *** symbol. **b** Comparison of top predictive tiles for the same metastatic FGFR3-mutant patient from the mUC cohort. The left slide shows the top predictive tiles on the primary site, while the right slide shows the top predictive tiles on the metastasis. For each group, we inspected the 50 most predictive tiles of the slide and sampled 6 representative tiles. Source data are provided as a Source Data file.

alterations in order to reduce testing costs and turnaround times. In case of poor RNA and DNA quality, such tools could further predict whether rebiopsy of metastasis or obtaining a liquid biopsy might be indicated.

Therefore, in the present study, we outlined the development and validation of a deep learning model trained for assessing the *FGFR3* mutation status of locally advanced and metastatic urothelial cancer patients by utilizing digitized H&E images of patient tumors. While we decided to develop an algorithm directly trained to predict the presence or absence of activating *FGFR3* mutations in H&E histopathology slides, other studies have demonstrated, that the presence of *FGFR* gene alterations can be predicted based on surrogate parameters such as low infiltration with tumor-infiltrating lymphocytes, which is supposed to be a common feature of *FGFR3* mutant urothelial carcinomas^{15,16}, but not a predictor for decreased responsivity to immunotherapy¹⁷. Validating these interesting findings from Velhamos et al.¹⁵, we also observed that top predictive tiles for the presence of *FGFR3* mutations were lymphocyte depleted¹⁵. Our results showed that the model identified at least 93% (up to 100%) of *FGFR3*-mutant cases in three independent validation cohorts. The model showcased

commendable AUCs (at least 82%) across three cohorts, reinforcing its robustness and clinical applicability. A paramount strength of our study lies in its extensive validation, covering a large spectrum of disease stages, from pT2 cases to metastatic patients, that were collected consecutively at multiple centers without specific inclusion or exclusion criteria. Notably, our sensitivity of 100% on the mUC cohort with a negative predictive value (NPV) of at least 98% underlines the clinical relevance of this assay to pre-screen patients who could benefit from an *FGFR3*-targeted therapy, like erdafitinib.

One notable aspect of our approach was including NMIBC cases in the training phase. Our strategy proved beneficial despite NMIBC and MIBC being distinct entities with different pathological and clinical pathways and morphology⁸. Although mixing NMIBC with MIBC/mUC is considered the “original sin” in bladder cancer research, integrating NMIBC cases likely introduced a broader spectrum of histological features associated with *FGFR3* mutations regardless of tumor stage, including monomorphic tumor cell appearance, low desmoplastic stroma content, and low or absent immune infiltration, enhancing the model’s generalizability. Although unconventional, this cross-disease training approach contributed significantly to the model’s impressive

performance in external validation. It underscores the potential of leveraging histopathological similarities across different stages of the disease to enhance an artificial intelligence model's performance. This finding also opens avenues for future research in developing more comprehensive models that span various cancer stages and types, potentially leading to more robust and universally applicable diagnostic tools. Furthermore, it underlines that features related to *FGFR3* mutations can be partially bound to specific morphological features maintained throughout progression and metastatic dissemination.

While our model performed well in detecting targetable *FGFR3* mutations, it is worth noting that in the early phase of the study, we also tried including *FGFR3* fusions in the MIBC I and TCGA cohort. Yet, our model could not learn to detect those alterations, most likely due to the rarity of these alterations (a total of 10 cases across $n = 543$ tumors of MIBC I and TCGA cohort). The scarcity of such cases in our dataset might have limited the model's ability to learn and recognize associated histological patterns. Furthermore, we hypothesized that associated patterns might be very different from tumors with activating *FGFR3* point mutations that cannot be sufficiently captured due to the very low number of cases. This remains speculation and has to be further elucidated in datasets with larger numbers of tumors with *FGFR3* fusions, which is important since patients with *FGFR3* fusions also benefit from erdafitinib treatment in metastatic stage (HR = 0.49, 95%-CI 0.23–1.03)¹⁰.

In conclusion, we demonstrated that H&E slides contain sufficient information to predict *FGFR3* mutations in muscle-invasive and metastatic urothelial cancer. These results pave the way for developing a diagnostic tool for detecting *FGFR3* mutations. While our results are promising, they must be confirmed through a large-scale, blind validation study to ensure the tool's reliability in a clinical setting. Crucially, our study calls for a subsequent calibration study to investigate how different operating thresholds might impact the predictive values of the model—ideally in patient cohorts treated with *FGFR3* inhibitors, that allow correlations with therapy outcomes. This calibration process is essential to adapt a potential diagnostic tool to the specific variations and conditions encountered in diverse clinical environments. In addition, further studies are required to study whether *FGFR3* fusions, that indicate *FGFR3* inhibition susceptibility, can be predicted from H&E histopathology slides.

Methods

Ethics approval

The present study was approved by the Institutional Ethical Review Board of Friedrich-Alexander-University Erlangen-Nuremberg FAU (protocol code 4607, 2015; protocol code 97_18Bc, 2018; protocol code 22-343-B, 2022; and protocol code 217_18C, 2018), and patients gave written informed consent. All experiments were carried out under the Declaration of Helsinki.

Datasets description

In total, we included $n = 1222$ patients with non-invasive or stroma-invasive (papillary) urothelial carcinomas (NMIBC: $n = 264$), muscle-invasive urothelial carcinomas (MIBC: $n = 440$; TCGA BLCA cohort: $n = 412$) and metastatic urothelial carcinomas (mUC, $n = 106$) of the bladder (Supplementary Table 1). Patients from all utilized cohorts were included consecutively in the sequence of treatment mentioned below to account for expected sex distribution for urothelial bladder cancer. As outlined in Supplementary Table 1 this strategy led to a balanced biological sex distribution (sex assigned at birth) as expected for urothelial (bladder) cancer. No self-reported gender data are available for the present cohorts. No inclusion or selection criteria were applied except the availability of FFPE tissue material for nucleic acid isolation and H&E stained tumor tissue carrying slides for digitization. Per sample a FFPE block with at least 30% tumor content (related to all tissue covered on the block) and at least an area of vital

non-necrotic 0.5×0.5 cm tumor tissue area with surrounding tumor stroma was selected by an experienced board-certified uropathologist (M.E.).

The MIBC cohort consists of $n = 546$ patients diagnosed with muscle-invasive urothelial carcinoma of the bladder who have been treated in curative intent by radical cystectomy plus lymphadenectomy and perioperative platinum-based chemotherapy according to applying EAU guidelines between 2008 and 2022 (Fig. 1a). Digitized H&E slides in conjunction with molecular data on *FGFR3* mutations were available in $n = 440$ patients of this cohort (split into two cohorts for training [MIBC I] and validation [MIBC II]). The NMIBC I cohort comprises $n = 155$ pTa and pT1 carcinoma patients treated by local surgical (transurethral resection) $\pm$ instillation therapy adapted to EAU risk classes between 2010 and 2020. The NMIBC II cohort consisted of $n = 109$ independent pTa carcinomas that were treated by transurethral resection followed by surveillance adapted to EAU risk classes between 2020–2023. The mUC cohort consists of 106 patients with metastatic urothelial cancer treated with standard regimes such as platinum-based chemotherapy and immunotherapy. All digitized H&E slides used in the present study were of intrinsic cancer tissues retrieved from the initial cancer diagnosis made on tissue material of the first transurethral resection or cystectomy without prior neoadjuvant chemotherapy treatment or instillation therapy. The *FGFR3* alteration status in these tumors was assessed by SNaPshot PCR as described in detail in the following sections. All cohorts were then scanned on the same slide scanner with identical settings. Thus, the present study cohort contains digitized H&E images that have been stained with a variety of H&E staining protocols in- and outside of Erlangen University Hospital, thus covering a significant variance in staining.

TCGA-BLCA is a multicentric cohort of 412 cases from the public TCGA database¹⁸. TCGA slides were used in digital format (downloaded from TCGA slide archive) and were H&E stained outside of our institution. Cases without a formalin-fixed paraffin-embedded tissue slide, *FGFR3* mutation status, and RNA sequencing were first filtered out. Slides of the remaining cases were then reviewed by an expert pathologist (M.E.), and cases without a slide containing tumor tissue covering MIBC were excluded. Ultimately, this led to the inclusion of 307 cases from the TCGA-BLCA repository (Fig. 1b and Supplementary Fig. 2). In our study, we denote this cohort TCGA. The *FGFR3* mutation status was obtained via the CBioPortal in the TCGA BLCA PanCan Atlas 2018 collection¹⁸. The cases harboring non-activating or inactivating *FGFR3* mutations/alterations (H349D, S344Y, P358L, V306I, L88Wfs*10, Q674*, E216K, D222N, G235D, and H791Tfs*29) were considered “wild-type” tumors as those mutations are not leading to a *FGFR* inhibitor targetable phenotype. The RNA sequencing was obtained from <https://portal.gdc.cancer.gov/>. Of these 397 cases, 24 were *FGFR3* mutants, corresponding to 8% of the cohort. Supplementary Table 7 provides a detailed breakdown of the most frequent mutations in all the cohorts. Supplementary Table 1 provides a detailed summary of the main clinical variables for all the cohorts included in the study.

Mutational detection via SNaPshot PCR and mRNA expression

DNA for mutational testing in NMIBC, MIBC, and mUC cohorts and RNA for gene expression profiling was isolated from FFPE using an automated procedure (Promega Maxwell, Promega, Wisconsin, USA). Five $10 \mu\text{m}$ FFPE sections with at least 50% tumor content were used per patient tumor and microdissected to maximize tumor purity. Sections were deparaffinized, tissues microdissected and suspended in $300 \mu\text{l}$ incubation buffer (Promega), and digested with Proteinase K (Promega) at 56°C overnight (550 rpm). DNA or RNA was then isolated from lysates using the Promega DNA FFPE purification Kit (Promega, Wisconsin, USA) according to the manufacturer's instructions. Afterward, DNA or RNA samples were purified, quantified, and quality-controlled using the Qubit Fluorometer (ThermoFisher).

FGFR3 mutation analysis was performed by a modified SNaPshot PCR protocol^{19–21}. Three regions of the *FGFR3* gene on Exons 7, 10, and 15 are amplified in a multiplex PCR followed by extension of mutation-specific primers with labeled dideoxynucleotides. After removing excess primers and dNTPs, eight SNaPshot primers detecting nine *FGFR3* mutations were annealed to the PCR products and extended with a labeled dideoxynucleotide. These extended primers were analyzed on an automatic sequencer, with the label on the incorporated nucleotide indicating the presence or absence of a mutation. A second and independent SNaPshot analysis verified all mutations. This allows sensitive detection of a total of eleven known activating *FGFR3* mutations that account for over 99% of known activating *FGFR3* mutations (p.R248C, p.S249C, p.G372C, p.G382R, p.S373C, p.Y375C, p.A393E, p.K652E, p.K652Q, p.K652M and p.K652T) and all known activating *FGFR3* mutations found in urothelial bladder cancer.

mRNA sequencing for *FGFR3* mRNA expression quantification in the MIBC I cohort was performed from 500 ng purified input RNA using the QuantSeq FWD Kit (Lexogen), which generates Illumina-compatible libraries from poly-adenylated mRNA^{22,23}. Sequencing was performed on an Illumina NovaSeq 6000 platform (single end; 1 × 75 bp), yielding at least 20 M clusters per sample. Preprocessing of demultiplexed raw data (FASTQ files) was performed via the nf-core RNA-seq pipeline v.3.3^{22,23}. TPM normalized expression data was log2 transformed for further analyses.

Preprocessing of whole-slide images

Whole slide images (WSI) are typically 100,000 × 100,000 pixels in dimension, making it computationally intractable to feed them directly to a deep learning model. We used a preprocessing pipeline to reduce the dimensionality of the data, which can be decomposed into three steps: 1) matter detection, 2) tiling, and 3) feature extraction. First, the tissue is detected on the WSI: a U-Net neural network segments part of the image that contains matter and discards regions with folded tissue, with artifacts (e.g. pen markers) or blurriness. This U-Net was trained on 460 H&E and IHC slides from an internal dataset where the tissue was manually annotated and validated on 115 slides with a Dice score of 0.96. Second, the regions detected as matter are tiled into smaller images (the “tiles”) of size 224 × 224 pixels, corresponding to an effective area of 224 × 224 μm (1.0 MPP). To be selected, a tile from an H&E slide must have at least 60% of the matter detected by the U-Net²⁴. Eventually, 1539 features are extracted from each selected tile using Bioptimus’ H0 (a Vision Transformer Giant)²⁵. The feature extractor for H&E slides is trained in a self-supervised fashion using DINOv2²⁶ without using any labels. For training and inference, the feature extractor weights were frozen. At the end of this preprocessing pipeline, each slide is represented by a matrix of size (n_tiles, 1539).

FGFR3 mutation prediction

The model detecting *FGFR3* mutations was trained on a discovery set comprising two cohorts (MIBC I, *n* = 239; NMIBC I, *n* = 155) and validated on three independent MIBC validation cohorts (TCGA, *n* = 373; MIBC II, *n* = 183; mUC, *n* = 96). The *FGFR3* mutation status was obtained by DNA sequencing using the SNaPshot method, described in detail in the “Datasets description” and “Mutational detection via SNaPshot PCR and mRNA expression sections” above²¹. The slides were split into small tiles of 224 × 224 pixels, corresponding to a tissue size of 224 × 224 μm at 1.0 MPP. The features were extracted with a pre-trained model from these tiles (more details in the Preprocessing of whole-slide images section).

The final model is an ensemble of 125 multiple-instance learning (MIL) models. The predictions of each model were aggregated by averaging all the output logits. Each MIL model uses the WSI preprocessing pipeline described in the section “Preprocessing of whole-slide images.” For training, a maximum of 5000 tiles are uniformly sampled from each slide for speed and memory considerations. All

models have an identical structure, as proposed by Courtiol et al.²⁷. First, each tile’s score corresponding to the *FGFR3* mutation status is computed. These scores are obtained from a multi-layer perceptron (MLP) with 128 hidden neurons followed by one neuron and a ReLU activation. We select the 100 top and bottom scores. Finally, these extreme scores are passed to a linear activation to obtain a final output logit. The logit is passed in a sigmoid layer to obtain a probability of whether the case is *FGFR3*-mutant or wild-type. The model was trained with the binary cross-entropy loss, using the labels obtained from the DNA sequencing described in the “DNA sequencing and *FGFR3* mutation status” section.

Performance assessment and statistical methods

The area under the receiver operating characteristic was used to quantify the model’s capability to distinguish between wild-type and *FGFR3*-mutant tumors. The sensitivity, specificity, PPV, and negative predictive value (NPV) to *FGFR3* mutations were used to assess the ability of the model to identify *FGFR3*-mutant cases. Confidence intervals at 95% confidence level were obtained by bootstrapping experiment results with 1000 repeats. All tests were two-tailed, and P-values < 0.05 were considered statistically significant. The intraclass correlation coefficient used to assess the model’s agreement between different slide predictions correspond to Pingouin’s ICC2k metric (a random sample of *k* judges rate each target).

Interpretability analysis

An expert pathologist (M.E.) reviewed 400 tiles predictive of *FGFR3* mutant status (*n* = 200) and wild-type status (*n* = 200) of TCGA. To analyze the morphological differences of the predictive tiles depending on the *FGFR3* expression level, the 200 predictive tiles of each status were selected to include 100 tiles with high expression and 100 tiles with low expression. The pathologist assessed twenty histological criteria (see Supplementary Methods) blinded to their model scores.

Uncertainty quantification and conformal prediction

To assess the reliability of our model’s predictions, we leveraged the ensemble nature of our prediction model by using a Deep Ensemble uncertainty quantification technique²⁸. The model comprises 125 individual multiple-instance learning models. For each patient, we calculated the variance of the predicted probabilities of the predicted class across these 125 models. This variance serves as a measure of uncertainty in the model’s predictions, providing insight into the confidence level of each classification.

To generate conformalized predictions^{29,30}, we split our external validation cohort into two equal groups: a calibration set and a test set. First, we calculated non-conformity scores for the calibration set, defining them as 1 – the predicted probability of the true class (either wild-type or *FGFR3*-mutant). Next, we determined a confidence threshold by taking the 95th percentile of these non-conformity scores. This allowed us to derive the prediction sets on the test set³¹. To assess our model confidence in its predictions, we monitored the expected cardinality of the prediction sets on the test set.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All images and the associated *FGFR3* mutational status for the TCGA cohort used in this study are publicly available at <https://portal.gdc.cancer.gov/> and cBioPortal (<https://www.cbioportal.org/>). The cohorts MIBC I, MIBC II, NMIBC I, NMIBC II, and mUC are the property of University Hospital Erlangen & Friedrich-Alexander-Universität Erlangen-Nürnberg. Image data, patient data and molecular data of patient cohorts of the University Hospital Erlangen (UKER) fall under

the European general data protection regulation (GDPR) and are available under restricted access given their potential sensitive nature. Access can be obtained for academic non-commercial research projects by contacting the corresponding authors. The processing time for data access requests may take up to one calendar month. Access to the sensitive data is restricted to the project duration specified by the requesting academic non-commercial researcher/research group. Source data are provided as a Source Data file. Source data are provided with this paper.

Code availability

An implementation of the U-Net is available at <https://github.com/milesial/Pytorch-UNet>. An implementation of BiOptimus H0 is available at <https://huggingface.co/bioptimus/H-optimus-0>. The packaged models can be found here: <https://github.com/PABannier/fgfr3mut> (<https://doi.org/10.5281/zenodo.13959977>)³², under a Creative Commons Attribution 4.0 International (CC-BY-4.0) License. The repositories referenced above contain license files and details of usage permissions. Any reuse of third-party software, including the U-Net implementation and BiOptimus H0, complies with their respective licenses.

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Author contributions

Study conception and design: M.E., C.S., C.Maussion., and P.-A.B.; data collection: P.M., C.Matek., N.K., J.B., R.W., D.S., B.S.-D., B.W., A.H., S.F., M.E.; software: P.-A.B., C.S.; analysis and interpretation of results: P.-A.B., C.S., M.E., M.T.; Draft manuscript preparation: P.-A.B., C.S., M.E. All authors reviewed the results and approved the final version of the manuscript.

Competing interests

Owkin employees own the company's stocks (P.-A.B., C.S., M.T., P.M., C.M.). B.S.-D: consultant for Arquer Ltd, UK, Cepheid, CA, USA, Concile GmbH, Germany, and Nucleix Inc., Israel. B.W.: received speaker's honoraria from MSD and Janssen-Cilag. N.K.: Personal fees, travel costs, and speaker's honoraria from Astellas, Novartis, Ipsen, Photocure, MSD, BicycleTX; advisory role for BicycleTX; research funding from BicycleTX. A.H.: Honoraria for lectures for Abbvie, AstraZeneca, Biontech, BMS, Boehringer Ingelheim, Cepheid, Diaceutics, Ipsen, Janssen, Lilly, MSD, Nanostring, Novartis, Roche, 3DHistotech; Consulting/advisory role for Abbvie, AstraZeneca, Biontech, BMS, Boehringer Ingelheim, Cepheid, Diaceutics, Gilead, Illumina, Ipsen, Janssen, Lilly, MSD, Nanostring, Novartis, Qiagen, QUIP GmbH, Roche, 3DHistotech; research support from AstraZeneca, Biontech, Cepheid, Gilead, Illumina, Janssen, Nanostring, Qiagen, QUIP GmbH, Roche. M.E.: Personal fees, travel costs, and speaker's honoraria from Eisai, MSD, AstraZeneca, Janssen-Cilag, Cepheid, Roche, Astellas, Diaceutics, Owkin, BMS, BicycleTX; research funding from AstraZeneca, Janssen-Cilag, STRATIFYER, Cepheid, Roche, Gilead, Owkin, QUIP GmbH, BicycleTX; advisory roles for Ferring, Diaceutics, MSD, AstraZeneca, Janssen-Cilag, GenomicHealth, Owkin, BMS, BicycleTX. R.M.W. is the CEO of STRATIFYER. All other authors declare no conflict of interest regarding the present work.

Additional information

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